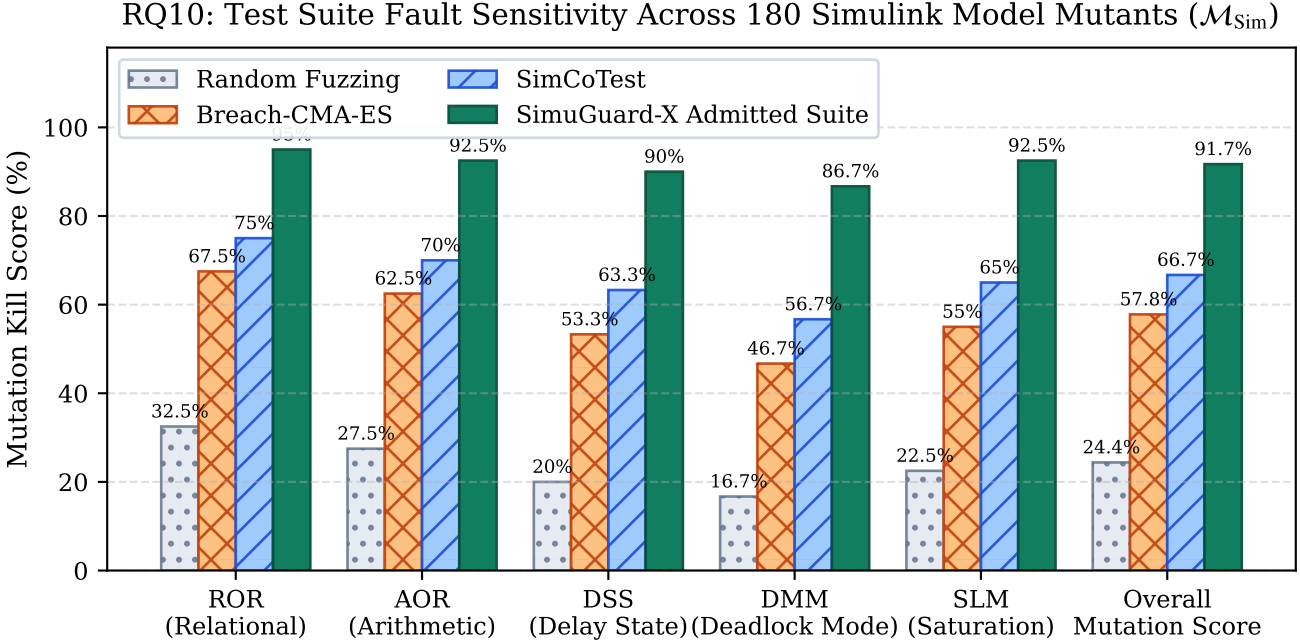
RQ10: Test Suite Fault Sensitivity Across 180 Simulink Model Mutants ($\mathcal{M}_{\text{Sim}}$)
Random Fuzzing
Breach-CMA-ES
SimCoTest
SimuGuard-X Admitted Suite
Mutation Kill Score (%)
0
20
40
60
80
100
32.5%
67.5%
75%
95%
27.5%
62.5%
70%
92.5%
20%
53.3%
63.3%
90%
16.7%
46.7%
56.7%
86.7%
22.5%
55%
65%
92.5%
24.4%
57.8%
66.7%
91.7%
ROR (Relational)
AOR (Arithmetic)
DSS (Delay State)
DMM (Deadlock Mode)
SLM (Saturation)
Overall Mutation Score